

Cell Design Specification

Basic specification

- System: Chen2020 baseline parameter set; architecture candidate TinySeparatorParticles.
- Contract targets: energy density ≥ 446.18 Wh/kg; 5C retention $\geq 90\%$; cell mass ≤ 0.040 kg.
- Result: ****not achieved****.

Performance verification

Metric	Value	Source	Result
Energy density	427.476639 Wh/kg	cell/v5_derived.json	FAIL
Cell mass	0.043216827 kg	cell/v5_derived.json	FAIL
5C retention	0.335044	cell/v5_derived.json	FAIL

Architecture parameters: separator 1 um; positive/negative particle radius 1 um. Electrolyte excluded fro